

Homework 10

Solution

1. Let f be the function defined by $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n(2n+1)}$ for every $-1 \leq x \leq 1$.

(a) Prove that f is well-defined.

Solution:

We show that the series $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n(2n+1)}$ convergence for every $-1 \leq x \leq 1$.

We note that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2n(2n+1)} = x \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{2n(2n+1)}$$

We have

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{2n(2n+1)} \underset{t=x^2}{=} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} t^n}{2n(2n+1)}$$

We compute

$$R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(-1)^{n+1}}{2n(2n+1)} \right|}} = 1$$

Thus the series converges for every $|x^2| < 1 \Rightarrow -1 < x < 1$.

We note that for $x = 1, -1$ we get a Leibniz series (or minus of a Leibniz series), thus the series

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cdot x^{2n+1}}{2n \cdot (2n+1)}$ converges for every $-1 \leq x \leq 1$.

Thus, f is well-defined.

(b) Find an explicit expression for f for every $-1 \leq x \leq 1$.

Solution:

We know that for every $-1 < x \leq 1$ we have

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n}$$

We substitute $x = t^2$ and divide both sides by 2, yielding

$$\frac{\ln(1+t^2)}{2} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} t^{2n}}{2n}$$

Thus for every $-1 < x < 1$ we have

$$\frac{\ln(1+x^2)}{2} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{2n}$$

Thus by term by term derivative for every $-1 < x < 1$ we have

$$f'(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^{2n}}{2n} = \frac{\ln(1+x^2)}{2}$$

thus there exists an $C \in \mathbb{R}$ such that

$$f(x) + C = \int \frac{\ln(1+x^2)}{2} dx = \frac{x \ln(1+x^2)}{2} - x + \arctan x$$

We note that

$$f(0) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cdot 0^{2n+1}}{2n \cdot (2n+1)} = 0$$

We substitute $x = 0$ and we get that

$$f(0) + C = \frac{0 \cdot \ln(1+0^2)}{2} - 0 + \arctan 0 \Rightarrow C = 0$$

Thus for every $-1 < x < 1$ we have

$$f(x) = \frac{x \ln(1+x^2)}{2} - x + \arctan x$$

For $x = 1$ we get that

$$f(1) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 1^{2n+1}}{2n(2n+1)} = \lim_{x \rightarrow 1^-} f(x) \stackrel{\text{continuous}}{=} f(1) = \frac{1 \cdot \ln(1+1^2)}{2} - 1 + \arctan(1) = \frac{\ln 2}{2} - 1 + \frac{\pi}{4}$$

For $x = -1$ we get that

$$f(1) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (-1)^{2n+1}}{2n(2n+1)} = \lim_{x \rightarrow -1^+} f(x) \stackrel{\text{continuous}}{=} f(-1) = \frac{(-1) \cdot \ln(1+(-1)^2)}{2} - (-1) + \arctan(-1) =$$

$$-\frac{\ln 2}{2} + 1 - \frac{\pi}{4}$$

Thus for every $-1 \leq x \leq 1$ we have

$$f(x) = \frac{x \ln(1+x^2)}{2} - x + \arctan x$$

(c) Conclude the value of the alternating series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n \cdot (2n+1)} = \frac{1}{2 \cdot 3} - \frac{1}{4 \cdot 5} + \frac{1}{6 \cdot 7} - \dots$$

Solution:

On the one hand, we substitute $x = 1$ and we get

$$f(1) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cdot 1^{2n+1}}{2n \cdot (2n+1)} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n \cdot (2n+1)} = \frac{1}{2 \cdot 3} - \frac{1}{4 \cdot 5} + \frac{1}{6 \cdot 7} - \dots$$

On the other hand

$$f(1) = \frac{1 \cdot \ln(1+1^2)}{2} - 1 + \arctan 1 = \frac{\ln 2}{2} - 1 + \frac{\pi}{4}$$

thus

$$\frac{1}{2 \cdot 3} - \frac{1}{4 \cdot 5} + \frac{1}{6 \cdot 7} - \dots = \frac{\ln 2}{2} - 1 + \frac{\pi}{4}$$

2. Consider the series $\sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n+1}}$, where $x \neq 1$.

(a) Find the domain of convergence of the series.

Solution:

We note that

$$\sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n+1}} = \frac{1}{x-1} \sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n}}$$

We consider

$$\sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n}} \stackrel{t=\frac{1}{(x-1)^2}}{=} \sum_{n=0}^{\infty} \frac{n+1}{2^n} t^n$$

We compute

$$R = \frac{1}{\lim_{n \rightarrow \infty} \sqrt[n]{\frac{n+1}{2^n}}} \stackrel{(\star)}{=} 2$$

($\star$) We note that for every $n \in \mathbb{N}$ we have

$$1 \leq \sqrt[n]{n+1} \leq \sqrt[n]{n+n} = 2^{\frac{1}{n}} n^{\frac{1}{n}}$$

As $\lim_{n \rightarrow \infty} 2^{\frac{1}{n}} = 1$ and $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$, then by squeeze rule we get that $\lim_{n \rightarrow \infty} \sqrt[n]{n+1} = 1$.

Thus, the series converges for every

$$\left| \frac{1}{(x-1)^2} \right| < 2 \Leftrightarrow x > 1 + \frac{1}{\sqrt{2}} \text{ or } x < 1 - \frac{1}{\sqrt{2}}$$

We note that for $x = 1 + \frac{1}{\sqrt{2}}$, $1 - \frac{1}{\sqrt{2}}$ the series diverges, since we get the series

$$\sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot \left(1 \pm \frac{1}{\sqrt{2}} - 1\right)^{2n}} = \sum_{n=0}^{\infty} (n+1)$$

and the necessary criterion does not hold.

Thus the domain of convergence is $x > 1 + \frac{1}{\sqrt{2}}$ or $x < 1 - \frac{1}{\sqrt{2}}$.

- (b) We define $f(x) = \sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n+1}}$ for every x in the domain of convergence found in item (a).

Find an explicit expression for f .

Solution:

We note that

$$\sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n+1}} = \frac{1}{x-1} \cdot \sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n}}$$

We have

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n}} &= \sum_{n=0}^{\infty} \frac{n+1}{\left(2(x-1)^2\right)^n} \stackrel{t=\frac{1}{2(x-1)^2}}{=} \sum_{n=0}^{\infty} [(n+1)t^n] = \sum_{n=0}^{\infty} (t^{n+1})' = \\ \left(\sum_{n=0}^{\infty} t^{n+1}\right)' &= \left(t \sum_{n=0}^{\infty} t^n\right)' = \left(t \cdot \frac{1}{1-t}\right)' = \left(\frac{t}{1-t}\right)' = \frac{1 \cdot (1-t) - t \cdot (-1)}{(1-t)^2} = \frac{1}{(1-t)^2} \end{aligned}$$

Thus we get that for every $x > 1 + \frac{1}{\sqrt{2}}$ or $x < 1 - \frac{1}{\sqrt{2}}$ we have

$$f(x) = \sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (x-1)^{2n+1}} = \frac{1}{x-1} \cdot \frac{1}{\left(1 - \frac{1}{2(x-1)^2}\right)^2}$$

- (c) Conclude the value of the series $\sum_{n=0}^{\infty} \frac{n+1}{2 \cdot 8^n}$.

Solution:

On the one hand, we substitute $x = 3 > 1 + \frac{1}{\sqrt{2}}$ and we get

$$f(3) = \sum_{n=0}^{\infty} \frac{n+1}{2^n \cdot (3-1)^{2n+1}} = \sum_{n=0}^{\infty} \frac{n+1}{2 \cdot 8^n}$$

On the other hand

$$f(3) = \frac{1}{3-1} \cdot \frac{1}{\left(1 - \frac{1}{2(3-1)^2}\right)^2} = \frac{32}{49}$$

Thus

$$\sum_{n=1}^{\infty} \frac{n+1}{2 \cdot 8^n} = \frac{32}{49}$$

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3. For items (a) and (b) develop each of the following functions into a power series around $x_0 = 0$, and find its domain of convergence.

(a) $\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt, \forall x \in \mathbb{R}.$

Solution:

First, we write the power series of $\sin t$, for every $t \in \mathbb{R}$ we have

$$\sin t = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot t^{2n+1}$$

Thus

$$\begin{cases} \frac{\sin(t)}{t} & t \neq 0 \\ 1 & t = 0 \end{cases} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot t^{2n}, \quad \forall t \in \mathbb{R}$$

Integrating term by term, we get

$$\text{Si}(x) = \int_0^x \frac{\sin t}{t} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot \int_0^x t^{2n} dt = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1) \cdot (2n+1)!} \cdot x^{2n+1} \quad \forall x \in \mathbb{R}$$

Consequently, the domain of convergence is $\mathbb{R}$.

- (b) $f(x) = \sinh x, \forall x \in \mathbb{R}$, where the hyperbolic sine function is defined by $\sinh x = \frac{e^x - e^{-x}}{2}$.

Solution:

We know that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \forall x \in \mathbb{R}$, thus

$$e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}, \quad \forall x \in \mathbb{R}$$

Subtracting the two series together, then for every $x \in \mathbb{R}$ we have

$$\begin{aligned} \sinh x &= \frac{e^x - e^{-x}}{2} = \frac{\sum_{n=0}^{\infty} \frac{x^n}{n!} - \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}}{2} = \sum_{n=0}^{\infty} \frac{(1 - (-1)^n)}{2 \cdot n!} \cdot x^n \\ &= \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} \cdot x^{2n+1} \end{aligned}$$

Consequently, the domain of convergence is $\mathbb{R}$.

(c) Consider the function defined by $f(x) = \ln(\sqrt{1-x^2})$, $\forall -1 < x < 1$.

i. Develop the function f into a power series around $x_0 = 0$ for every $-1 < x < 1$.

Solution:

Let $-1 < x < 1$ and let $t = -x^2$. Then $-1 < t < 0$ and therefore we can write

$$\begin{aligned} f(x) &= \ln(\sqrt{1-x^2}) = \ln\left[(1-x^2)^{\frac{1}{2}}\right] = \frac{1}{2} \ln(1-x^2) \\ &\stackrel{t=-x^2}{=} \frac{1}{2} \ln(1+t) = \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} t^n}{n} = \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (-x^2)^n}{n} \\ &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (-1)^n x^{2n}}{n} = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{x^{2n}}{n} \end{aligned}$$

ii. Let $\alpha = \int_0^{\sqrt{\frac{1}{2}}} f(x) dx$. Compute the value of the following series as a function of α :

$$\frac{1}{2 \cdot 1 \cdot 3} + \frac{1}{2^2 \cdot 2 \cdot 5} + \frac{1}{2^3 \cdot 3 \cdot 7} + \frac{1}{2^4 \cdot 4 \cdot 9} + \frac{1}{2^5 \cdot 5 \cdot 11} + \dots$$

Solution:

We note that

$$\begin{aligned} \alpha &= \int_0^{\sqrt{\frac{1}{2}}} f(x) dx = \int_0^{\sqrt{\frac{1}{2}}} \left[-\frac{1}{2} \sum_{n=1}^{\infty} \frac{x^{2n}}{n} \right] dx = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} \int_0^{\sqrt{\frac{1}{2}}} x^{2n} dx \\ &= -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} \left[\frac{x^{2n+1}}{2n+1} \right]_0^{\sqrt{\frac{1}{2}}} = -\frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} \frac{\left(\sqrt{\frac{1}{2}}\right)^{2n+1}}{2n+1} = -\frac{1}{2} \cdot \sqrt{\frac{1}{2}} \sum_{n=1}^{\infty} \frac{1}{2^n \cdot n \cdot (2n+1)} \end{aligned}$$

Thus

$$\frac{1}{2 \cdot 1 \cdot 3} + \frac{1}{2^2 \cdot 2 \cdot 5} + \frac{1}{2^3 \cdot 3 \cdot 7} + \frac{1}{2^4 \cdot 4 \cdot 9} + \frac{1}{2^5 \cdot 5 \cdot 11} + \dots = \sum_{n=1}^{\infty} \frac{1}{2^n \cdot n \cdot (2n+1)} = -2\sqrt{2}\alpha$$

4. (a) Find the Taylor series about $x_0 = 0$ for the function defined by $f(x) = \begin{cases} \frac{\arctan(x)}{x} & x \neq 0 \\ 1 & x = 0 \end{cases}$ for every $x \in \mathbb{R}$, and find its domain of convergence.

Solution:

We know that the power series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$ converges if and only if $-1 \leq x \leq 1$, and that for every $-1 \leq x \leq 1$ we have

$$\arctan(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}$$

It follows that for every $-1 \leq x \leq 1$ such that $x \neq 0$ we have

$$f(x) = \frac{\arctan(x)}{x} = \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$$

Note that for $x = 0$ it holds that

$$f(0) = 1 = 1 - 0 + 0 - 0 + \dots$$

and therefore $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$ if and only if $-1 \leq x \leq 1$.

By the theorem of uniqueness of power series, we conclude that the Taylor series of f is

$$T_f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots$$

and the domain of convergence is $[-1, 1]$.

- (b) Find $f^{(n)}(0)$ for every $0 \leq n \in \mathbb{Z}$.

Solution:

As

$$f(x) = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k}$$

then

$$a_n = \begin{cases} 0 & n = 2k - 1 \\ \frac{(-1)^k}{2k+1} & n = 2k \end{cases}$$

thus by the theorem of uniqueness of power series, for every $0 \leq n \in \mathbb{Z}$ we have

$$f^{(n)}(0) = a_n \cdot n! = \begin{cases} 0 & n = 2k - 1 \\ \frac{(2k)!(-1)^k}{2k+1} & n = 2k \end{cases}$$